file.cio

Purpose

`file.cio` is the master index. SWAT+ opens this file first. Each line tells the model which file to read for one input category. Every other input file in the project is reached through `file.cio`.

Source

- Reader: src/readcio_read.f90
- File-name fields and defaults: src/input_file_module.f90

Format

- Line 1: title line. Skipped by the reader.
- Lines 2 to 31: one line per input category. Each line starts with a category label (a free-form token that the reader reads but discards), followed by a fixed number of file-name fields whose order matches the corresponding `in_*` type in `input_file_module.f90`.
- Line 32: a single line for the optional output path.
- A field set to the literal token `null` disables the corresponding file.

The category labels written by tools like QSWAT+ and SWAT+ Editor are conventional. The reader does not validate them; only field order and count matter.

Categories and fields

The table lists every category in the order the reader expects them, the number of file-name fields, and the default file names defined in `input_file_module.f90`.

#	Category label	Fields (defaults from `input_file_module.f90`)
1	simulation	`time.sim`, `print.prt`, `object.prt`, `object.cnt`, `constituents.cs`
2	basin	`codes.bsn`, `parameters.bsn`
3	climate	`weather-sta.cli`, `weather-wgn.cli`, `pet.cli`, `pcp.cli`, `tmp.cli`, `slr.cli`, `hmd.cli`, `wnd.cli`, `atmodep.cli`
4	connect	`hru.con`, `hru-lte.con`, `rout_unit.con`, `gwflow.con`, `aquifer.con`, `aquifer2d.con`, `channel.con`, `reservoir.con`, `recall.con`, `exco.con`, `delratio.con`, `outlet.con`, `chandeg.con`
5	channel	`initial.cha`, `channel.cha`, `hydrology.cha`, `sediment.cha`, `nutrients.cha`, `channel-lte.cha`, `hyd-sed-lte.cha`, `temperature.cha`
6	reservoir	`initial.res`, `reservoir.res`, `hydrology.res`, `sediment.res`, `nutrients.res`, `weir.res`, `wetland.wet`, `hydrology.wet`
7	routing_unit	`rout_unit.def`, `rout_unit.ele`, `rout_unit.rtu`, `rout_unit.dr`
8	hru	`hru-data.hru`, `hru-lte.hru`

#	Category label	Fields (defaults from `input_file_module.f90`)
9	exco	`exco.exc`, `exco_om.exc`, `exco_pest.exc`, `exco_path.exc`, `exco_hmet.exc`, `exco_salt.exc`
10	recall	`recall.rec`
11	dr	`delratio.del`, `dr_om.del`, `dr_pest.del`, `dr_path.del`, `dr_hmet.del`, `dr_salt.del`
12	aquifer	`initial.aqu`, `aquifer.aqu`
13	herd	`animal.hrd`, `herd.hrd`, `ranch.hrd`
14	water_rights	`water_allocation.wro`, `element.wro`, `water_rights.wro`
15	link	`chan-surf.lin`, `aqu_cha.lin`
16	hydrology	`hydrology.hyd`, `topography.hyd`, `field.fld`
17	structural	`tiledrain.str`, `septic.str`, `filterstrip.str`, `grassedww.str`, `bmpuser.str`
18	hru_parm_db	`plants.plt`, `fertilizer.frt`, `tillage.til`, `pesticide.pes`, `pathogens.pth`, `metals.mtl`, `salt.slt`, `urban.urb`, `septic.sep`, `snow.sno`
19	ops	`harv.ops`, `graze.ops`, `irr.ops`, `chem_app.ops`, `fire.ops`, `sweep.ops`
20	lum	`landuse.lum`, `management.sch`, `cntable.lum`, `cons_practice.lum`, `ovn_table.lum`
21	chg	`cal_parms.cal`, `calibration.cal`, `codes.sft`, `wb_parms.sft`, `water_balance.sft`, `ch_sed_budget.sft`, `ch_sed_parms.sft`, `plant_parms.sft`, `plant_gro.sft`
22	init	`plant.ini`, `soil_plant.ini`, `om_water.ini`, `pest_hru.ini`, `pest_water.ini`, `path_hru.ini`, `path_water.ini`, `hmet_hru.ini`, `hmet_water.ini`, `salt_hru.ini`, `salt_water.ini`
23	soils	`soils.sol`, `nutrients.sol`, `soils_lte.sol`
24	decision_table	`lum.dtl`, `res_rel.dtl`, `scen_lu.dtl`, `flo_con.dtl`
25	regions	`ls_unit.ele`, `ls_unit.def`, `ls_reg.ele`, `ls_reg.def`, `ls_cal.reg`, `ch_catunit.ele`, `ch_catunit.def`, `ch_reg.def`, `aqu_catunit.ele`, `aqu_catunit.def`, `aqu_reg.def`, `res_catunit.ele`, `res_catunit.def`, `res_reg.def`, `rec_catunit.ele`, `rec_catunit.def`, `rec_reg.def`
26	pcp_path	one path string
27	tmp_path	one path string
28	slr_path	one path string
29	hmd_path	one path string
30	wnd_path	one path string
31	output_path	one path string (read as the whole line after the label)

Categories 1 to 30 are read with list-directed I/O. Category 31 (`output_path`) is read with format `(A)` and the value is taken from the trailing portion of the line after the first space.

Example

Top of `refdata/Ames_sub1/file.cio`:

```
file.cio: AMES
```

```
simulation      time.sim          print.prt        null             object.cnt       null
basin           codes.bsn         parameters.bsn
climate         weather-sta.cli   weather-wgn.cli  null             pcp.cli          tmp.cli
connect         hru.con           null             null             null             null
channel         null       null       null       null        null       null       null       null
reservoir       null       null       null       null        null       null       null       null
routing_unit    null       null       null       null
hru             hru-data.hru      null
exco            null              null               null             null             null
recall          null
dr              null              null               null             null             null
aquifer         nbull             null
```

Notes on the example:

- Line 2 (`simulation`): only `time.sim`, `print.prt`, and `object.cnt` are in use. Slots 3 and 5 (`object.prt`, `constituents.cs`) are set to `null`.
- Line 4 (`climate`): only stations, weather generator, precipitation, and temperature are active. PET, solar, humidity, wind, and atmospheric deposition are disabled.
- Line 14 (`aquifer`): the dataset writes `nbull` for the initial-aquifer file. This is the file name in this project, not a default and not a typo of `null`. The model opens whatever file name appears here.

Related files

Every input file referenced in `file.cio`. Start from the Input reference index.